1 Introduction

Perfusion MRI refers to techniques that visualize how blood flows into tissues following the injection of contrast material. This project simulates a simplified 2D perfusion MRI experiment using a synthetic phantom composed of three compartments: a blood vessel, normal tissue, and abnormal tissue with contrast retention.

The goal is not to model full MRI physics but to capture qualitative perfusion patterns such as:

- rapid wash-in and wash-out in a vessel
- moderate contrast uptake in normal tissue
- slower uptake and prolonged retention in abnormal tissue (common in tumors).

This project was chosen because it provides an intuitive connection between computer science concepts (arrays, image formation, signal curves) and how clinicians interpret perfusion changes that can indicate disease.

Aims

- Aim 1: Create a synthetic 2D phantom with vessel, normal tissue, and abnormal tissue regions.
- Aim 2: Simulate contrast dynamics using mathematical perfusion models.
- Aim 3: Generate output images, SI–time curves, and perfusion maps that visualize differences between tissue types.
- Aim 4: Build a GUI to allow interactive parameter control and phantom preview.
- Aim 5: Analyze results by comparing output images to the phantom ground truth.

2 Methods

This section describes the software environment, phantom construction, perfusion model, GUI, and outputs.

2.1 Software and Libraries

The project is implemented in Python using:

- NumPy - array creation, masks, kinetic models
- Matplotlib - plotting and GIF animation

- Tkinter - GUI
- Pillow - GIF writing
- JSON / CSV - saving SI curves and simulation parameters

The GUI allows changing the matrix size, tissue kinetics, time steps, etc.

2.2 Phantom Description (Ground Truth)

The phantom is a 256×256 image divided into three regions: Central vessel: a vertical stripe of adjustable width Normal tissue: left side Abnormal tissue: right side

Mask construction uses simple coordinate-based thresholding.

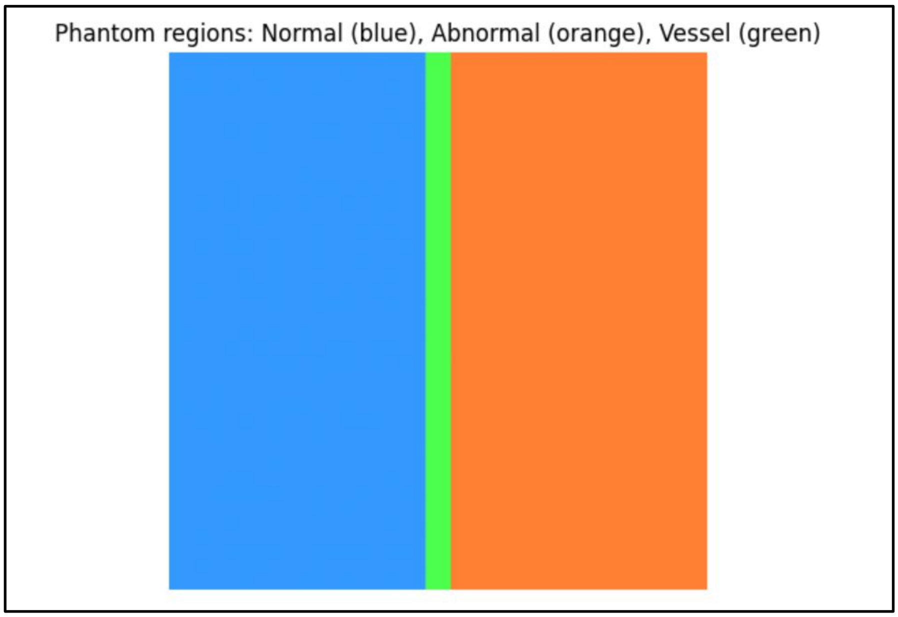


Figure 1. Phantom layout showing normal (blue), abnormal (orange), and vessel (green) tissue regions. This defines the ground truth.

2.3 Perfusion Model (Equations)

Each region has a baseline signal intensity (SI = 1.0) and receives an added contrast-dependent term modeled using exponentials:

Vessel (bolus):

$$\Delta SI_{vessel}(t) = A \cdot e^{-\frac{\max(0,\, t_{\mathrm{tp}} - t)}{t_{\mathrm{tp}}/3}} \cdot e^{-\frac{\max(0,\, t - t_{\mathrm{tp}})}{\tau_{\mathrm{wash}}}}$$

Normal Tissue:

$$\Delta SI_{\text{normal}}(t) = a\left(1 - e^{-t/\tau_{\text{rise}}}\right) e^{-t/\tau_{\text{decay}}}$$

Abnormal Tissue (retention):

$$\Delta SI_{\text{abnormal}}(t) = a\left(1 - e^{-t/\tau_{\text{rise}}}\right) e^{-t/\tau_{\text{decay,long}}}$$

Abnormal tissue has:

- slower rise
- slower decay
- higher late retention

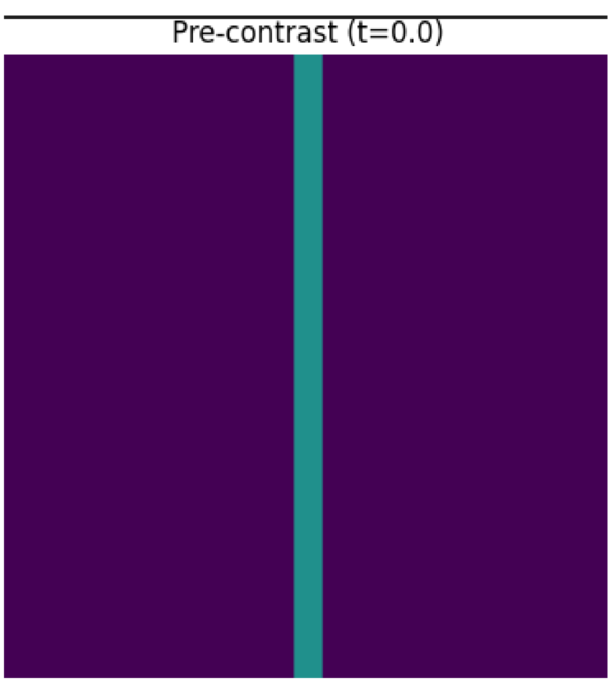
This simulates tumor-like perfusion behavior.

2.4 Frame Generation

For each time point:

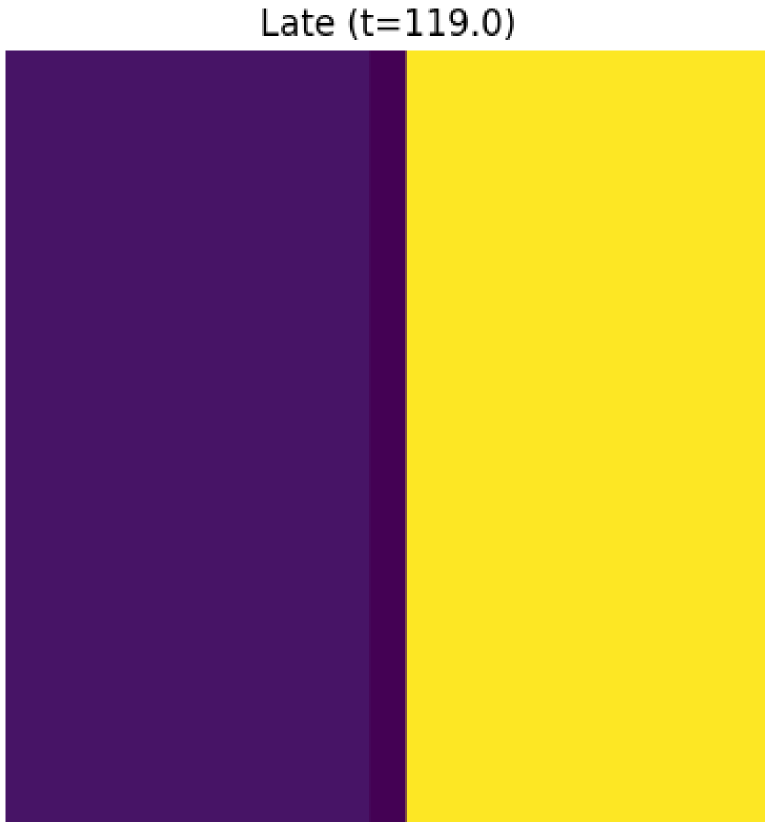
- compute region SI values
- assign values using masks
- assemble a 2D image
- save ~120 images for the GIF

Pre-contrast frame (frame_pre.png)



Time-to-peak frame (frame_ttp.png)

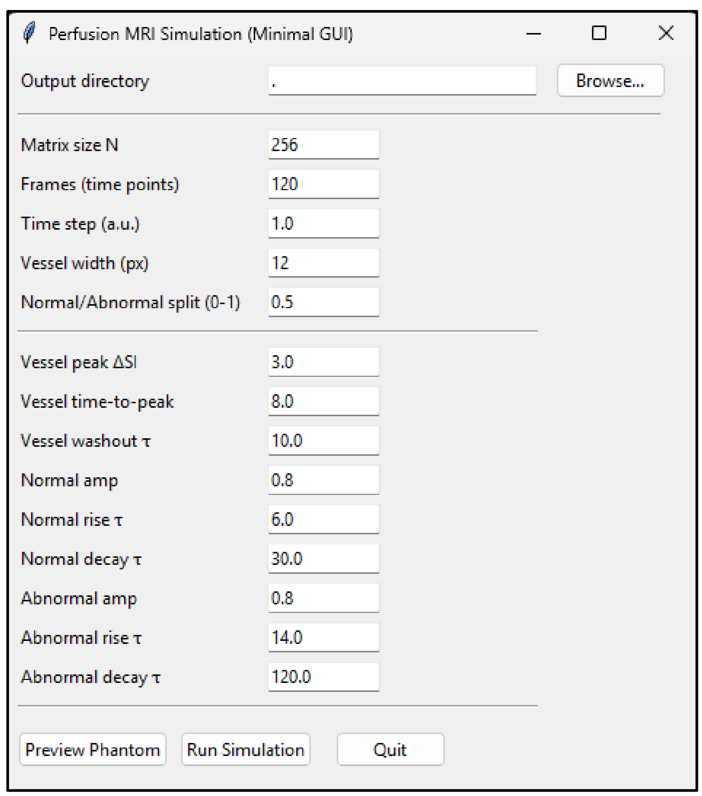
Late frame (frame_late.png)



3.5 GUI Implementation

A Tkinter window organizes parameters into:

- Phantom / Geometry
  - (matrix size, vessel width, split)
- Perfusion Kinetics
  - (amplitudes, rise τ, decay τ)
- Controls
  - (Preview Phantom, Run, etc)



2.6 Outputs Generated

When a simulation is run, the program produces the following:

- perfusion_sim.gif
- si_curves.png, si_curves.csv
- si_stats.csv, si_stats.txt
- frame_pre.png, frame_ttp.png, frame_late.png
- delta_map.png
- params.json

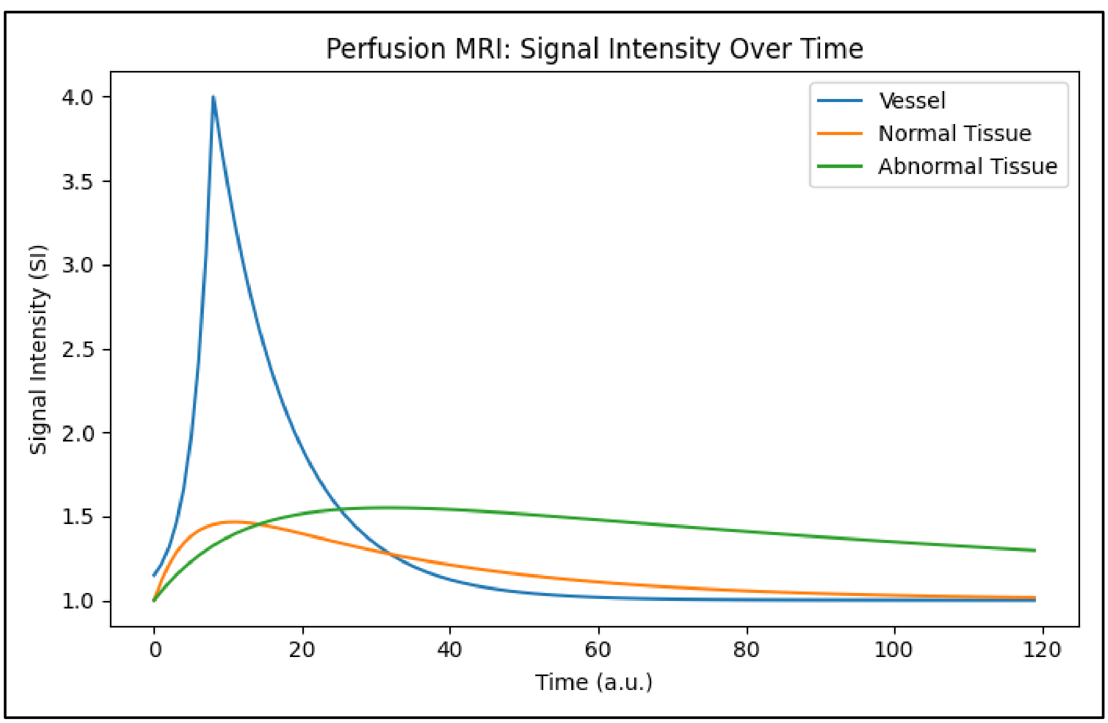
3 Results and Discussion

Results correspond to Aims 2-5.

3.1 Perfusion Curves (Aim 2)

- Vessel peaks sharply at the bolus arrival, then decays quickly.
- Normal tissue shows moderate uptake and washout.
- Abnormal tissue shows slower rise and noticeably higher late intensity.

This matches expected tumor perfusion behavior.



3.2 Quantitative Statistics (Aim 2–3)

Tables are automatically generated in si_stats.csv. Summaries include:

- Vessel: highest Peak SI and earliest TTP
- Normal: moderate AUC
- Abnormal: largest baseline-subtracted AUC and highest Late-SI

These values confirm that abnormal tissue retains contrast longer.

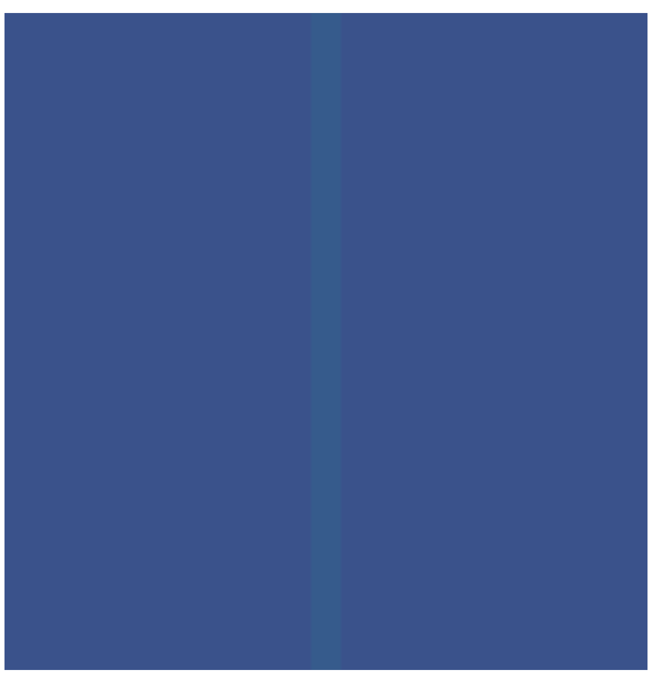
Region	Peak SI	Time-to-P	AUC	AUC(bsl-s	Late_SI(la	Mean	Std
Vessel	4	8	156.71511	37.715119	1.0000740	1.3149151	0.6177016188525734
Normal Ti	1.4657899	11	138.53438	19.53438	1.0176815	1.1628496	0.1451585961534065
Abnormal	1.5504258	32	169.35397	50.353979	1.3081058	1.4208527	0.10755805252915528

si_stats

3.3 Perfusion Animation (Aim 3)

The vessel lights up first, normal tissue next, and abnormal tissue retains contrast longer.

Perfusion MRI Simulation



3.4 SI Map (Aim 3 + Ground Truth Comparison)

Interpretation:

- The abnormal region shows highest ΔSI (late - pre).
- Vessel shows little ΔSI late because it washes out.
- Normal tissue returns closer to baseline.

This matches the ground truth phantom exactly, confirming the simulation behaves as designed.

4 Conclusions

This project successfully achieved all aims:

- A well-defined phantom was constructed.
- Physiological perfusion behaviors were simulated using simple exponential models.
- A GUI was built to allow parameter control and phantom preview.
- Outputs included curves, images, GIF animation, and statistical metrics.
- Analysis confirmed that the abnormal tissue retained contrast longer, as expected clinically.

The project demonstrates how computational simulation can mimic functional medical imaging and help visualize perfusion dynamics.